Do Simple Physics Priors Change Empirical Scaling in Scientific Machine Learning?

A Controlled Lotka–Volterra Case Study

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Abstract

Physics-informed learning methods are often evaluated at a small number of fixed model and dataset sizes, making it difficult to determine whether physical priors change learning behavior in a systematic way or merely improve performance in isolated settings. We present a controlled study that reframes this question through empirical scaling laws. Using fixed-horizon flow-map prediction on the Lotka–Volterra system, we compare a plain multilayer perceptron (MLP) against two physics-regularized variants—an ODE-residual prior and a conservation-law prior—across model capacity and dataset size. The scaling-law framework serves both as a performance summary and as a diagnostic tool: it reveals not only whether a prior helps, but whether it changes the shape of learning, and whether the prior itself is well-matched to the supervised task. We fit scaling relationships of the form

$$E(N, D) \approx E_\infty + aN^{-\alpha} + bD^{-\beta},$$

where E is test error, N is model capacity, and D is dataset size. The ODE-residual prior, which incurs an irreducible approximation error at the tested horizon, degrades scaling uniformly and raises the error floor by an order of magnitude. The conservation-law prior, which is exact, **[PENDING: conservation sweep results: state whether it improves data efficiency, changes exponents, or lowers error floor]**. These contrasting outcomes demonstrate that empirical scaling laws can distinguish a well-matched physics prior from a mismatched one, providing a principled diagnostic beyond point-wise benchmark comparisons.

1 Introduction

Scientific machine learning (SciML) aims to combine data-driven models with scientific structure such as governing equations, invariants, symmetries, and conservation laws [14, 6]. A central hope is that such structure improves data efficiency, robustness, and interpretability. However, comparisons between physics-informed and purely data-driven methods are often made at only a few fixed dataset sizes and model capacities. As a result, it remains unclear whether physics-informed priors change learning in a systematic way or simply shift performance in isolated experimental settings.

In mainstream machine learning, empirical scaling laws have become a useful lens for understanding how test error changes as a function of model size, dataset size, and compute [5, 4]. By contrast, comparable scaling-based evaluation is still underdeveloped in SciML. This is a notable gap because scientific learning problems are especially sensitive to regime: task difficulty depends not

only on sample count, but also on time horizon, parameter variability, discretization, optimization stability, and the form of the physical prior itself.

This paper studies a controlled instance of that broader question. We consider fixed-horizon flow-map prediction on the Lotka–Volterra system and ask:

> Does adding a simple physics prior measurably change empirical scaling behavior with respect to model capacity and dataset size?

Crucially, we study *two* physics priors on the same system and task: one that is an approximate proxy for the dynamics (a midpoint-rule ODE residual) and one that is an exact invariant (the Lotka–Volterra conservation law). This pairing allows us to go beyond "does the prior help?" and instead ask whether the scaling-law framework *diagnoses* the quality of the prior itself.

Our contributions are:

1. A minimal, reproducible scaling-law protocol for comparing data-only and physics-regularized models on a controlled dynamical system.
2. A two-axis scaling study over model capacity (N) and dataset size (D) for fixed-horizon flow-map prediction.
3. Empirical demonstration that scaling-law signatures distinguish a well-matched prior (exact conservation law) from a mismatched one (biased ODE residual), producing qualitatively different effects on exponents, error floors, and training stability.
4. A diagnostic methodology—complementing scaling fits with λ-sensitivity analysis and ground-truth residual checks—that can be reused in broader SciML evaluations.

2 Related Work

2.1 Physics-informed scientific machine learning

Physics-informed learning methods inject scientific structure into machine learning models in several ways. Soft approaches typically introduce residual penalties derived from governing equations [11, 3]. Harder approaches build physical structure directly into the architecture or parameterization [14, 2]. While these methods are widely studied, their evaluation is often benchmark-specific and restricted to a small number of scales.

2.2 Empirical scaling laws

Empirical scaling laws describe how performance changes as a function of data, model size, or compute. In natural language processing and vision, such laws have proven useful both descriptively and operationally [5, 4]. They provide a compact summary of learning efficiency and can reveal regime changes or diminishing returns. Recent work explains scaling phenomena through variance-limited and resolution-limited regimes [1], while constructive models can predict generalization error across scales [12].

2.3 Scaling as an evaluation lens for SciML

The use of scaling laws in SciML is still limited. Theoretical work on neural scaling for elliptic PDEs has begun to establish generalization bounds [9], but these results are confined to specific PDE classes. There remains a need for controlled empirical work asking whether physical priors change the shape of learning curves in a reproducible and interpretable way.

3 Problem Setting

3.1 Task formulation

We study supervised fixed-horizon flow-map prediction for the Lotka–Volterra system. Let $\boldsymbol{x}(t) = (u(t), v(t)) \in \mathbb{R}^2$ denote the system state. Given an initial condition $\boldsymbol{x}_0$ and a fixed prediction horizon $T > 0$, the learning task is to approximate the flow map

$$\Phi_T : \boldsymbol{x}_0 \mapsto \boldsymbol{x}(T).$$

This formulation provides a well-defined supervised target while avoiding the confounding effects of long autoregressive rollout.

3.2 Lotka–Volterra dynamics

The Lotka–Volterra system is defined by

$$\frac{du}{dt} = \alpha u - \beta uv, \tag{1}$$

$$\frac{dv}{dt} = \delta uv - \gamma v, \tag{2}$$

where u and v denote prey and predator populations, and $\alpha, \beta, \gamma, \delta > 0$ are system parameters. We use fixed parameters $(\alpha, \beta, \delta, \gamma) = (1.5, 1.0, 1.0, 3.0)$ throughout.

The system possesses an exact first integral (conservation law):

$$H(u, v) = \delta u - \gamma \ln u + \beta v - \alpha \ln v, \tag{3}$$

satisfying $H(\boldsymbol{x}(t)) = H(\boldsymbol{x}_0)$ for all t. This invariant plays a central role in our experimental design: it provides an *exact* physics prior that can be contrasted with an *approximate* one.

3.3 Data generation

Initial conditions are sampled uniformly from $[0.5, 2.5]^2$. Ground-truth flow maps are computed using the DOP853 solver (Dormand–Prince 8(5,3)) at tolerances `rtol` $= 10^{-9}$, `atol` $= 10^{-11}$, with prediction horizon $T = 1.0$. The dataset comprises 20,000 training pairs, 2,000 validation, and 2,000 test pairs. Training subsets of sizes $D \in \{64, 128, 256, 512, 1024, 2048, 4096, 8192\}$ are drawn as nested prefixes of the same shuffled training pool, ensuring that each smaller set is a subset of every larger one. Three independent data seeds (11, 22, 33) provide distinct pools.

4 Models

4.1 Plain MLP baseline

Our baseline is a standard multilayer perceptron (MLP) with ReLU activations:

$$\hat{\boldsymbol{x}}_T = f_\phi(\boldsymbol{x}_0),$$

trained to minimize the mean squared error

$$\mathcal{L}_{\text{data}} = \frac{1}{B} \sum_{i=1}^{B} \left\| f_\phi(\boldsymbol{x}_0^{(i)}) - \boldsymbol{x}_T^{(i)} \right\|_2^2.$$

Input and output are standardized using training-set statistics; the loss is computed in normalized space.

4.2 Capacity grid

Model capacity N (parameter count) is varied through a fixed architecture grid:

Name	Hidden widths	Parameters (N)
Tiny	$[32, 32]$	1,218
Small	$[64, 64]$	4,482
Medium	$[128, 128]$	17,154
Large	$[256, 256]$	67,074
XLarge	$[256, 256, 256]$	132,866

4.3 ODE-residual prior (midpoint)

The first physics-regularized model augments the data loss with a midpoint-rule ODE residual:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{mid}},$$

where

$$\mathcal{L}_{\text{mid}} = \frac{1}{B} \sum_{i=1}^{B} \left\| \hat{\boldsymbol{x}}_T^{(i)} - \boldsymbol{x}_0^{(i)} - T \cdot F\left(\frac{\boldsymbol{x}_0^{(i)} + \hat{\boldsymbol{x}}_T^{(i)}}{2} \right) \right\|_2^2$$

and F is the Lotka–Volterra vector field from Equations (1) and (2). This is a first-order implicit-midpoint quadrature approximation to the integral form of the ODE, computed on denormalized (physical) state variables.

4.4 Conservation-law prior

The second physics-regularized model penalizes violation of the exact first integral:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{cons}},$$

where

$$\mathcal{L}_{\text{cons}} = \frac{1}{B} \sum_{i=1}^{B} \left(H(\hat{\boldsymbol{x}}_T^{(i)}) - H(\boldsymbol{x}_0^{(i)}) \right)^2$$

and H is the invariant from Equation (3). Unlike the midpoint residual, this prior has zero irreducible error: $\mathcal{L}_{\text{cons}} = 0$ on exact flow-map targets.

5 Experimental Design

5.1 Scaling axes

We vary two quantities systematically:

- **Model capacity** N: five sizes from 1,218 to 132,866 parameters.
- **Dataset size** D: eight sizes from 64 to 8,192 training samples.

The use of nested subsets reduces confounding due to sample composition.

5.2 Evaluation metric

The primary metric is the mean relative ℓ_2 prediction error on the held-out test set, computed in physical (denormalized) coordinates:

$$E(N, D) = \frac{1}{M} \sum_{i=1}^{M} \frac{\left\| \hat{\boldsymbol{x}}_T^{(i)} - \boldsymbol{x}_T^{(i)} \right\|_2}{\left\| \boldsymbol{x}_T^{(i)} \right\|_2 + \varepsilon},$$

where $\varepsilon > 0$ is a small stability constant.

5.3 Scaling-law ansatz

We fit empirical relationships of the form

$$E(N, D) \approx E_\infty + aN^{-\alpha} + bD^{-\beta}, \tag{4}$$

where E_∞ is an asymptotic error floor, α is the capacity-scaling exponent, β is the data-scaling exponent, and $a, b > 0$ are fitted coefficients. The purpose is descriptive and comparative, not predictive.

5.4 Statistical protocol

Each (N, D) configuration is run across 3 data seeds $\times$ 3 training seeds $=$ 9 replicates. We report mean and standard error across replicates. Scaling-law parameters are estimated with 1,000 bootstrap samples to produce 95% confidence intervals.

5.5 Training details

All models use Adam with learning rate 10^{-3}, weight decay 10^{-6}, batch size $\min(128, D)$, and train for up to 400 epochs with early stopping (patience 40 on validation relative ℓ_2). Zero runs diverged or produced NaN across the entire experimental grid.

6 Research Questions

1. Does the physics prior improve prediction error, especially in the low-data regime?
2. Does the prior change the data-scaling exponent β?
3. Does the prior change the capacity-scaling exponent α?
4. Does the prior affect training stability or the optimization trajectory?
5. Can the scaling-law framework distinguish a well-matched prior from a mismatched one?

7 Results

7.1 Plain MLP baseline: scaling behavior

The plain MLP exhibits clean power-law scaling across both axes. Representative test errors (mean $\pm$ stderr over 9 replicates):

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Tiny ($N = 1218$)	0.0164 ± 0.0006	0.0055 ± 0.0002	0.0026 ± 0.0001	0.0017 ± 0.0001
Small ($N = 4482$)	0.0103 ± 0.0005	0.0030 ± 0.0001	0.0015 ± 0.0001	0.0010 ± 0.0000
Medium ($N = 17154$)	0.0070 ± 0.0007	0.0016 ± 0.0001	0.0010 ± 0.0000	0.0008 ± 0.0001
Large ($N = 67074$)	0.0049 ± 0.0005	0.0011 ± 0.0000	0.0009 ± 0.0000	0.0008 ± 0.0000
XLarge ($N = 132866$)	0.0045 ± 0.0004	0.0012 ± 0.0000	0.0009 ± 0.0000	0.0007 ± 0.0000

The combined scaling fit yields:

$$\hat{E}_\infty \approx 0, \quad \hat{\alpha} \approx 0.91, \quad \hat{\beta} \approx 0.85.$$

These exponents are stable across bootstrap resamples. The near-zero error floor indicates that adding more data and capacity continues to yield improvement across the tested range.

[Generate plots from runs-progress data: test error vs. D for each capacity, log–log axes, plain model only.]

Figure 1: Data scaling for the plain MLP baseline. Test error decreases as a power law with exponent $\hat{\beta} \approx 0.85$.

7.2 ODE-residual prior: biased scaling

The midpoint-residual PIML model (with $\lambda_{\text{phys}} = 0.1$) shows qualitatively different scaling behavior. Test errors plateau at approximately $E \approx 0.014$–0.016 across all (N, D) configurations, with weak dependence on either axis:

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Tiny	0.0244 ± 0.0006	0.0176 ± 0.0002	0.0162 ± 0.0002	0.0155 ± 0.0002
Small	0.0222 ± 0.0006	0.0170 ± 0.0002	0.0162 ± 0.0001	0.0155 ± 0.0002
Medium	0.0202 ± 0.0008	0.0163 ± 0.0001	0.0157 ± 0.0003	0.0148 ± 0.0003
Large	0.0198 ± 0.0007	0.0159 ± 0.0001	0.0150 ± 0.0005	0.0143 ± 0.0002
XLarge	0.0192 ± 0.0007	0.0154 ± 0.0002	0.0139 ± 0.0004	0.0141 ± 0.0002

The combined scaling fit yields:

$$\hat{E}_\infty \approx 0.011, \quad \hat{\alpha} \approx 0.16, \quad \hat{\beta} \approx 0.53.$$

The high error floor (20× the plain model's best error) and shallow exponents indicate that the prior dominates the optimization, preventing the model from exploiting additional data or capacity.

[Generate comparative plot: plain vs. midpoint PIML, test error vs. D, at fixed large capacity.]

Figure 2: Comparative data scaling for the plain and midpoint-residual models at large capacity. The PIML model saturates at the physics-loss floor.

Interpretation. The midpoint-rule residual is a first-order quadrature approximation. At the tested horizon $T = 1.0$, even ground-truth flow-map targets incur a mean relative residual of approximately 15% under this approximation. The physics loss therefore has an irreducible floor of $\mathcal{L}^*_{\text{mid}} \approx 0.15$, which is $\sim$50,000$\times$ larger than the achievable data loss ($\sim 3 \times 10^{-6}$). As training progresses and data loss decreases, the physics gradient dominates, and the optimizer converges to a plateau determined by the biased prior rather than the data.

7.3 Conservation-law prior: exact scaling

[PENDING: Run conservation-law lambda sweep and full scaling sweep.]
[PENDING: Fill in representative error table, same format as above.]
[PENDING: Fill in combined scaling fit: $\hat{E}_\infty$, $\hat{\alpha}$, $\hat{\beta}$.]

Expected contrast. Because the conservation-law prior has zero irreducible error on ground-truth targets (verified: $\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$ on the test set), we expect qualitatively different behavior from the midpoint prior. The key questions are whether the prior:

- lowers the error floor relative to the plain model,
- steepens the data-scaling exponent (improved data efficiency),
- changes capacity scaling, or
- primarily helps in the low-data regime.

7.4 Comparative scaling-law fits

Table 1: Fitted scaling parameters across the three models. The combined fit uses all (N, D) configurations.

Model	$\hat{E}_\infty$	$\hat{\alpha}$	$\hat{\beta}$	Notes
Plain MLP	≈ 0	0.91	0.85	Clean power-law scalin
PIML (midpoint)	≈ 0.011	0.16	0.53	Dominated by biased pr
PIML (conservation)	**[PENDING: –]**	**[PENDING: –]**	**[PENDING: –]**	**[PENDING: conservation**

7.5 Training stability

All 720 runs (360 plain + 360 midpoint PIML) completed without divergence or NaN across the full (N, D) grid. The midpoint PIML model early-stops substantially earlier: best-epoch values are typically 50–100 versus 300–500 for the plain model, consistent with the optimizer saturating at the physics-loss plateau.

[PENDING: Add conservation-law stability data.]

[Generate heatmap of best-epoch across (N, D) for each model.]

Figure 3: Training duration (best epoch) across the (N, D) grid. The midpoint PIML model early-stops much sooner due to the physics-loss plateau.

7.6 Sensitivity to regularization strength

We performed a dedicated λ_{phys} sweep for the midpoint-residual prior across $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ at two capacities (small, large) and three dataset sizes (128, 1024, 4096), with 9 replicates per configuration (324 runs total).

Table 2: Midpoint-prior λ_{phys} sweep: test relative ℓ_2 error (mean over 9 replicates).

	λ_{phys}					
Config	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1
Small, $D=128$	0.0067	0.0070	0.0068	0.0076	0.0194	0.0748
Small, $D=1024$	0.0021	0.0021	0.0021	0.0028	0.0165	0.0711
Small, $D=4096$	0.0013	0.0012	0.0012	0.0020	0.0158	0.0706
Large, $D=128$	0.0028	0.0027	0.0028	0.0035	0.0176	0.0598
Large, $D=1024$	0.0009	0.0008	0.0009	0.0019	0.0157	0.0630
Large, $D=4096$	0.0008	0.0008	0.0008	0.0015	0.0142	0.0687

Key observations:

- $\lambda \leq 10^{-3}$: indistinguishable from the plain baseline.
- $\lambda = 10^{-2}$: onset of degradation (1.5–2.2× worse).
- $\lambda = 10^{-1}$: strong degradation (8–20×); error collapses to ≈ 0.015 regardless of D or N.
- $\lambda = 1$: severe degradation (50–100×).
- No value of λ improves over the baseline at any configuration.

The $\lambda = 0$ path reproduces the plain model bit-for-bit, confirming correct implementation. The monotonic dose–response pattern rules out isolated hyperparameter failure and is consistent with a biased prior.

[PENDING: Add conservation-law lambda sweep results in same format.]

8 Discussion

8.1 Scaling signatures as prior diagnostics

The central finding so far is that the scaling-law framework provides qualitatively different signatures for the two priors tested:

- **Biased prior (midpoint residual):** elevated error floor ($\hat{E}_\infty \approx 0.011$), shallow exponents ($\hat{\alpha} \approx 0.16$, $\hat{\beta} \approx 0.53$), early stopping at a physics-dominated plateau. The scaling law detects a prior–task mismatch.
- **Exact prior (conservation law):** **[PENDING: fill in after conservation sweep]**.

This demonstrates that scaling laws are not merely a summarization tool—they can serve as a diagnostic for whether a physics prior is well-matched to the learning task. A biased prior does not simply fail to help; it actively distorts the scaling surface in a characteristic and interpretable way.

8.2 Why the midpoint prior fails

The midpoint-rule residual approximates $\int_0^T F(\boldsymbol{x}(t))\,dt$ with a single evaluation at the midpoint $(\boldsymbol{x}_0 + \boldsymbol{x}_T)/2$. At $T = 1.0$, the Lotka–Volterra trajectory curvature makes this approximation poor:

ground-truth targets incur $\sim$15% relative residual. As the data loss decreases during training, the physics gradient grows relative to the data gradient:

$$\frac{\lambda \cdot \nabla \mathcal{L}_{\text{mid}}}{\nabla \mathcal{L}_{\text{data}}} \sim \frac{\lambda \cdot O(1)}{O(\mathcal{L}_{\text{data}})},$$

eventually dominating the optimization. This mechanism explains both the observed error plateau and the premature early stopping.

8.3 Relationship to the broader research program

This paper is the first pilot in a broader research program to evaluate physics-informed priors via empirical scaling laws (see [1, 9] for related directions). The present work contributes the experimental methodology and a first calibration result on a minimal dynamical system. Future work will extend to:

- multiple ODE regimes and PDE flow-map tasks,
- different forms of physical priors (PINN-style collocation, architectural constraints, operator-learning),
- regime-aware analyses of where scaling laws hold or fail.

8.4 Why Lotka–Volterra?

Lotka–Volterra provides a minimal nonlinear dynamical system that is simple enough for exhaustive sweeps yet nontrivial enough to make inductive bias relevant. It possesses an exact first integral, enabling a controlled comparison between an exact and an approximate prior on the same task. Conclusions should not be overgeneralized to PDE learning, chaotic systems, or more complex physics-informed architectures.

9 Limitations

1. The study uses a single dynamical system (Lotka–Volterra).
2. It compares a plain MLP against two soft-constraint regularizers; architectural physics priors are not tested.
3. It focuses on fixed-horizon prediction ($T = 1.0$) rather than long autoregressive rollout.
4. It uses noise-free synthetic data.
5. The scaling-law ansatz Equation (4) is a descriptive approximation, not a theoretical prediction.
6. **[PENDING: If conservation results are null: note that the exact prior (conservation law) constrains only one scalar quantity, leaving the model underconstrained along the conserved manifold.]**

10 Conclusion

We introduced a controlled study for evaluating whether simple physics priors change empirical scaling behavior in scientific machine learning. Using fixed-horizon Lotka–Volterra flow-map prediction, we compared a plain MLP against an ODE-residual prior and a conservation-law prior across model capacity and dataset size. The scaling-law framework reveals qualitatively different signatures: the biased midpoint prior raises the error floor and flattens scaling exponents, while the exact conservation prior **[PENDING: state outcome]**.

The main contribution is methodological. Scaling laws serve not only as a compact summary of learning behavior, but as a diagnostic tool that distinguishes well-matched priors from mismatched ones. This perspective—treating scaling behavior itself as the object of study—supports a more principled and reproducible basis for evaluating physical inductive biases in scientific machine learning.

Acknowledgments

Acknowledgments placeholder.

A Physics-Loss Weight Diagnostic

A.1 Motivation

A natural concern in any negative result involving a regularized objective is whether the outcome is caused by an implementation bug, an ill-chosen λ_{phys}, or a genuine prior–task mismatch. This appendix documents a systematic diagnostic that distinguishes these cases.

A.2 Implementation equivalence control

With $\lambda_{\text{phys}} = 0$, the PIML code path reproduces the plain model bit-for-bit at all tested configurations (Table 2). This rules out hidden code-path discrepancies.

A.3 Ground-truth residual

Evaluating the midpoint residual on exact ground-truth targets:

$$\text{Mean } \|r\|_2 = 0.363, \qquad \text{Mean relative residual} \approx 15\%, \qquad \mathcal{L}^*_{\text{mid}} \approx 0.151.$$

The conservation-law residual on the same targets: $\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$.

A.4 Interpretation

The midpoint physics loss has an irreducible floor $\sim$50,000$\times$ larger than the achievable data loss. No value of λ can fix this: small λ makes the prior invisible; large λ drives the optimizer toward the biased proxy. This creates a characteristic dose–response degradation pattern (Table 2) that is a hallmark of prior–task mismatch.

[PENDING: Add equivalent lambda-sweep data for conservation prior.]

B Conservation-Law Prior: Details

The Lotka–Volterra first integral is

$$H(u, v) = \delta u - \gamma \ln u + \beta v - \alpha \ln v.$$

The conservation loss penalizes $(\Delta H)^2 = (H(\hat{\boldsymbol{x}}_T) - H(\boldsymbol{x}_0))^2$. On ground-truth data, $|\Delta H|/|H| \approx 10^{-7}$, confirming the prior is exact to within numerical integration tolerances.

[PENDING: Add architecture details, hyperparameter tables, and any additional ablation results from the conservation sweep.]

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